A2 Part 4 - Tasks 3, 4

Mason Stanley

Task 3

p_bubble_1.c: Each thread sorts its own chunk and checks the boundary with the next chunk. A barrier syncs threads after every pass.

Pros: simple, gets faster with more threads.

Cons: no early stop, barrier overhead, 1 thread is slower than sequential.

p_bubble_2.c: Odd-even sort. Each phase compares non-overlapping pairs so threads don't conflict. Barrier between phases.

Pros: safer, faster than sequential even with 1 thread.

Cons: still $O(n^2)$, adding threads past 2 barely helped.

Task 4

Program	Sequential	1 Thread	2 Threads	3 Threads	4 Threads
Num_Integrate (s)	1.814868	1.826168	0.914919	0.616706	0.467724
Merge_approach_1 (ms)	14.842	15.181	6.962	5.345	7.381
Merge_approach_2 (ms)	14.842	14.269	6.970	7.198	7.180
Merge_approach_3 (ms)	14.842	15.690	7.062	5.990	5.237
Bubble_1 (s)	16.596022	19.878040	13.843831	10.777551	9.770916
Bubble_2 (s)	16.596022	11.009647	6.409503	6.236547	6.117009

N = 1,000,000,000 for integration, SIZE = 100,000 for sorts. Prints commented out when timing.